

K.BALAJI

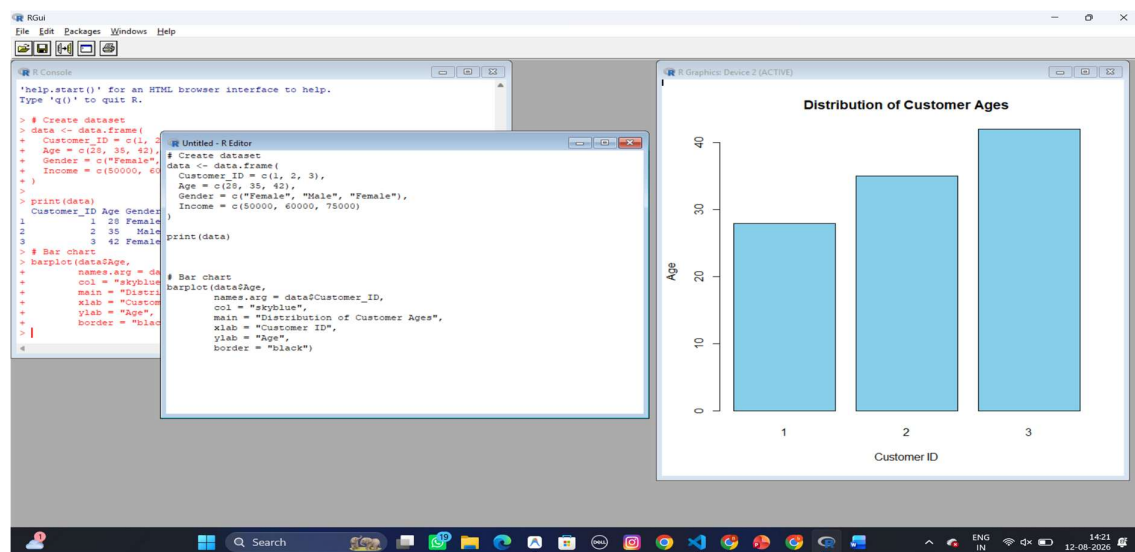
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DATA HANDLING AND VISUVALIZATION LAB 16-20

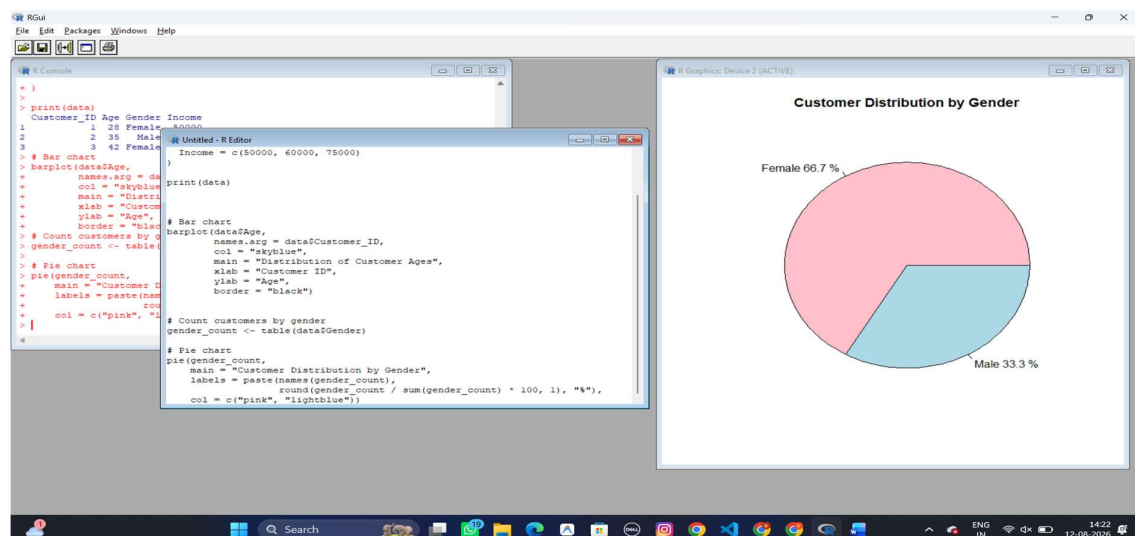
16) Customer Demographics Analysis

Dataset: Customer Demographics

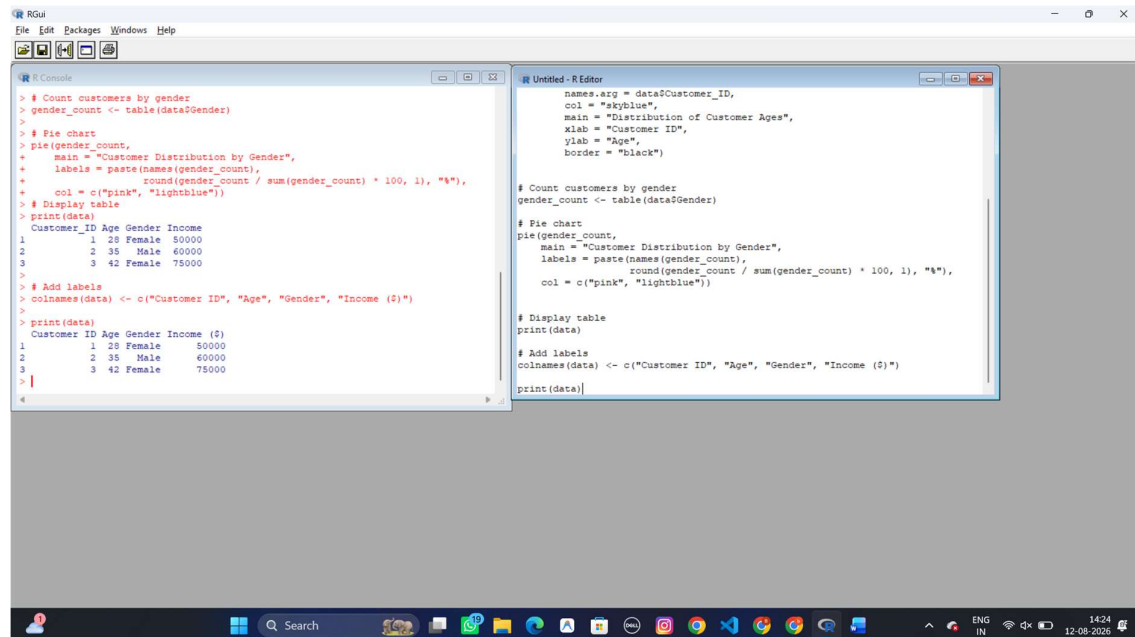
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.



2. Generate a pie chart to represent the distribution of customers by gender.



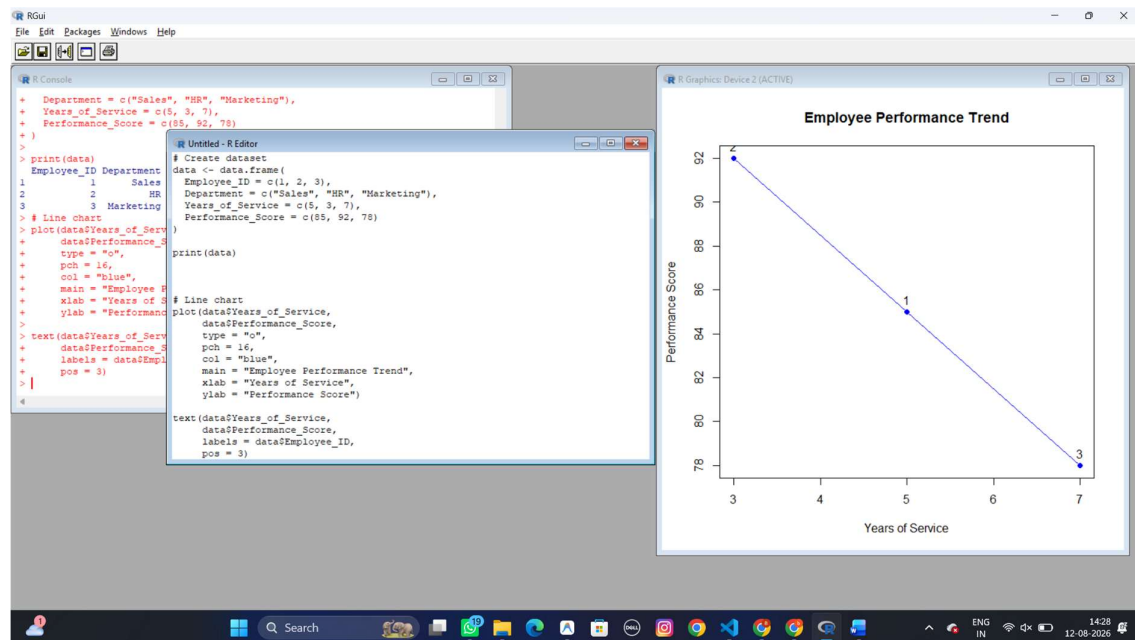
3. Build a table to show the demographic information for each customer. Label the table elements.



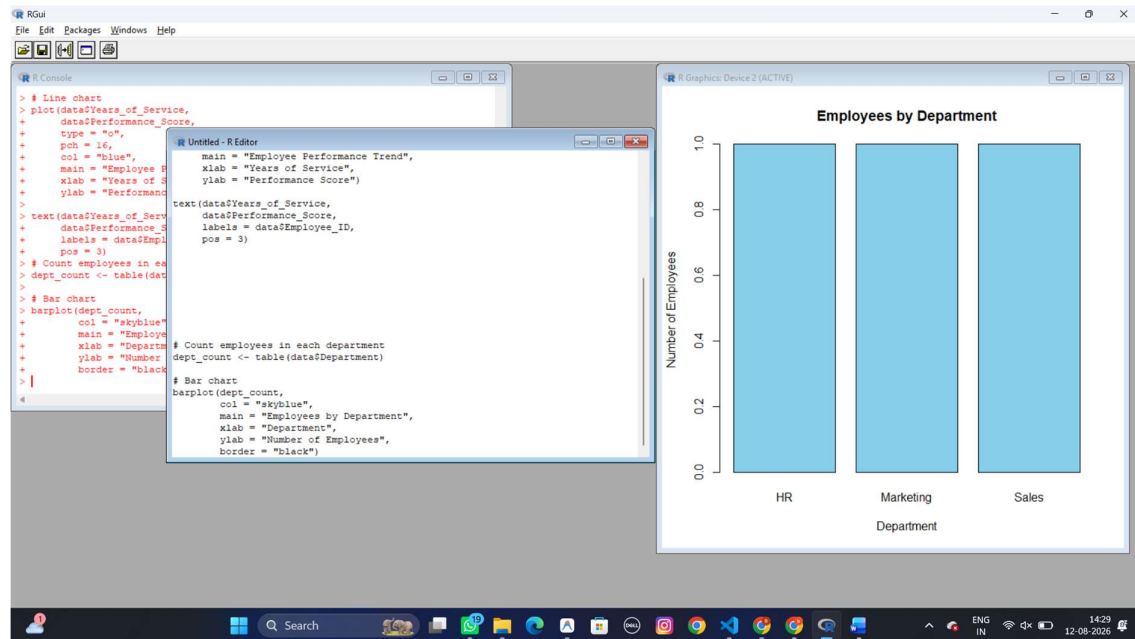
17: Employee Performance Analysis

Dataset: Employee Performance

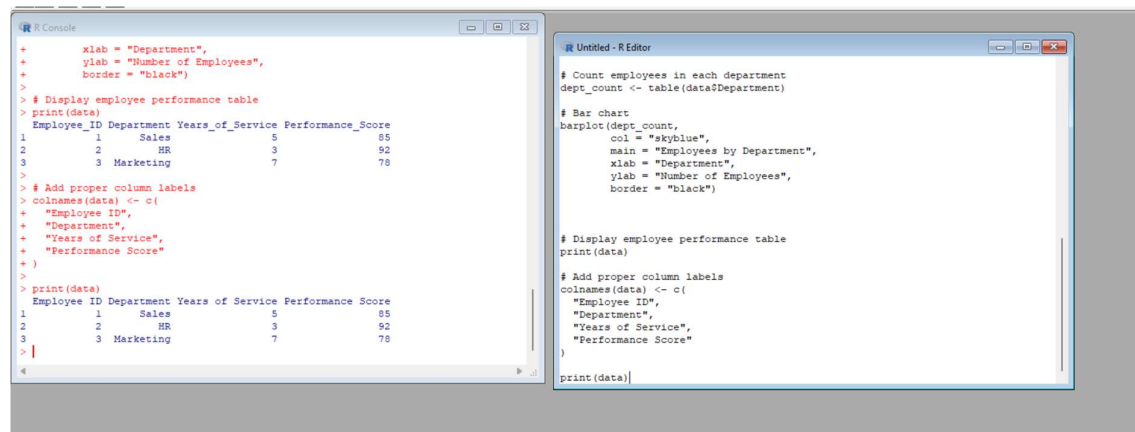
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.



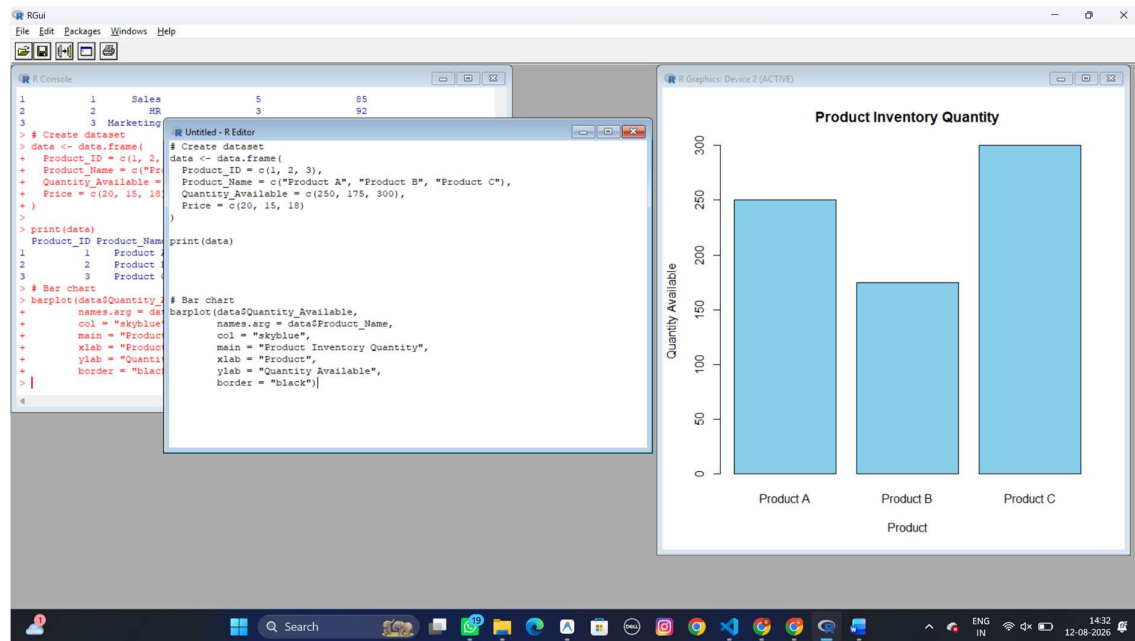
3. Build a table to display the performance data for each employee. Label the table elements.



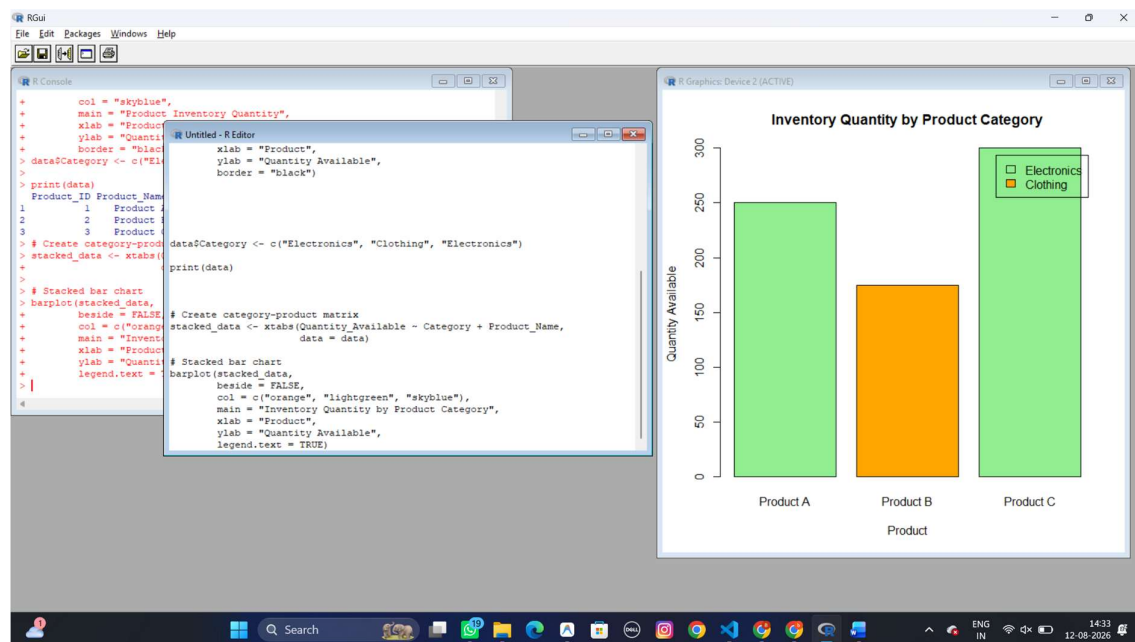
18: Product Inventory Management

Dataset: Product Inventory

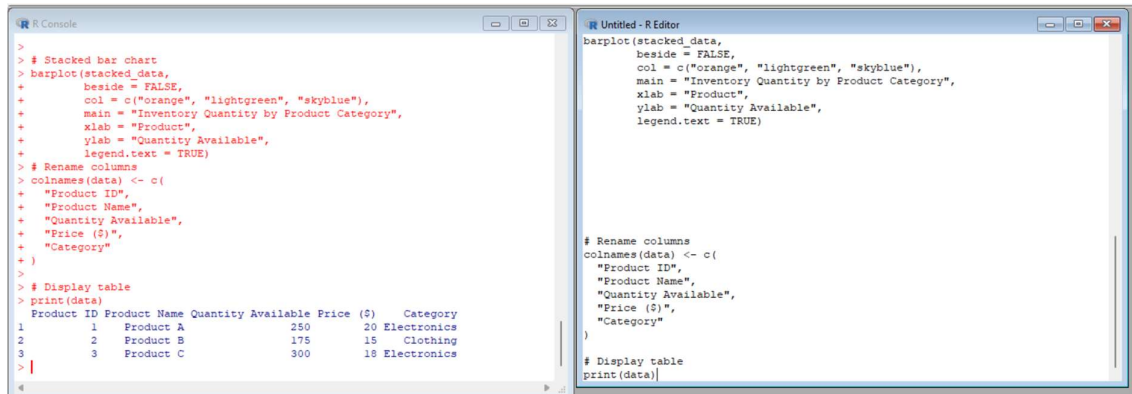
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.



2. Generate a stacked bar chart to show the quantity of each product within different product categories.



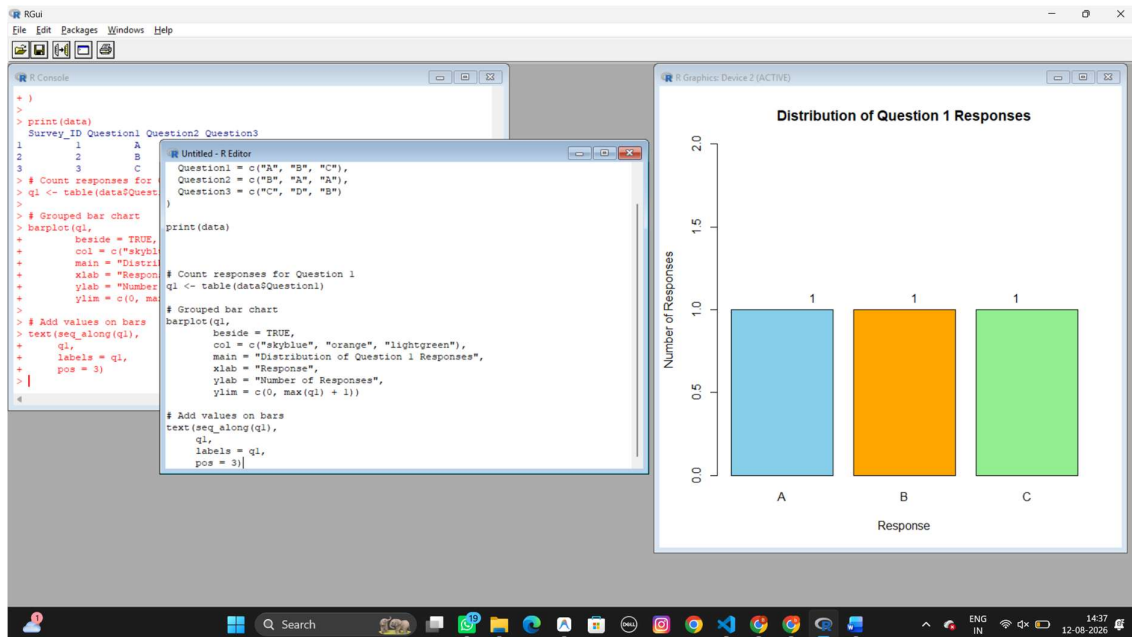
3. Build a table to show the inventory data for each product. Label the table elements.



Set 19: Survey Responses Analysis

Dataset: Survey Responses

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```

R Console
> response_matrix <- matrix(
+   0,
+   nrow = length(responses),
+   ncol = 3
+ )
>
> rownames(response_matrix) <- responses
> colnames(response_matrix) <- c(
+   "Question 1",
+   "Question 2",
+   "Question 3"
+ )
>
> # Fill counts
> response_matrix[names(q1), "Question 1"] <- q1
> response_matrix[names(q2), "Question 2"] <- q2
> response_matrix[names(q3), "Question 3"] <- q3
>
> print(response_matrix)
  Question 1 Question 2 Question 3
A          1          2          0
B          1          1          1
C          1          0          1
D          0          0          1
>
>

Untitled - R Editor
responses <- sort(unique(c(
  data$Question1,
  data$Question2,
  data$Question3
)))

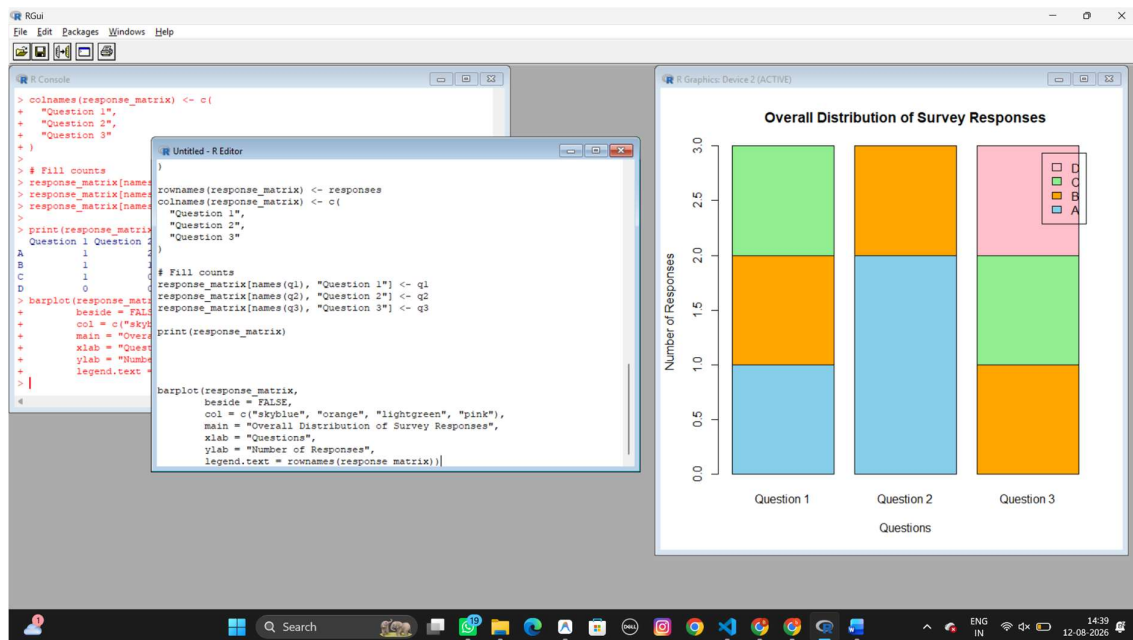
# Create matrix
response_matrix <- matrix(
  0,
  nrow = length(responses),
  ncol = 3
)

rownames(response_matrix) <- responses
colnames(response_matrix) <- c(
  "Question 1",
  "Question 2",
  "Question 3"
)

# Fill counts
response_matrix[names(q1), "Question 1"] <- q1
response_matrix[names(q2), "Question 2"] <- q2
response_matrix[names(q3), "Question 3"] <- q3

print(response_matrix)

```



3. Build a table to show the survey response data for each survey. Label the table elements.

```

R Console
B          1          1          1
C          1          0          1
D          0          0          1
>
> barplot(response_matrix,
+   beside = FALSE,
+   col = c("skyblue", "orange", "lightgreen", "pink"),
+   main = "Overall Distribution of Survey Responses",
+   xlab = "Questions",
+   ylab = "Number of Responses",
+   legend.text = rownames(response_matrix))
>
> # Rename columns
> colnames(data) <- c(
+   "Survey ID",
+   "Question 1",
+   "Question 2",
+   "Question 3"
+ )
>
> # Display table
> print(data)
  Survey ID Question 1 Question 2 Question 3
1          1          A          B          C
2          2          B          A          D
3          3          C          A          B
>
>

Untitled - R Editor
barplot(response_matrix,
  beside = FALSE,
  col = c("skyblue", "orange", "lightgreen", "pink"),
  main = "Overall Distribution of Survey Responses",
  xlab = "Questions",
  ylab = "Number of Responses",
  legend.text = rownames(response_matrix))

# Rename columns
colnames(data) <- c(
  "Survey ID",
  "Question 1",
  "Question 2",
  "Question 3"
)

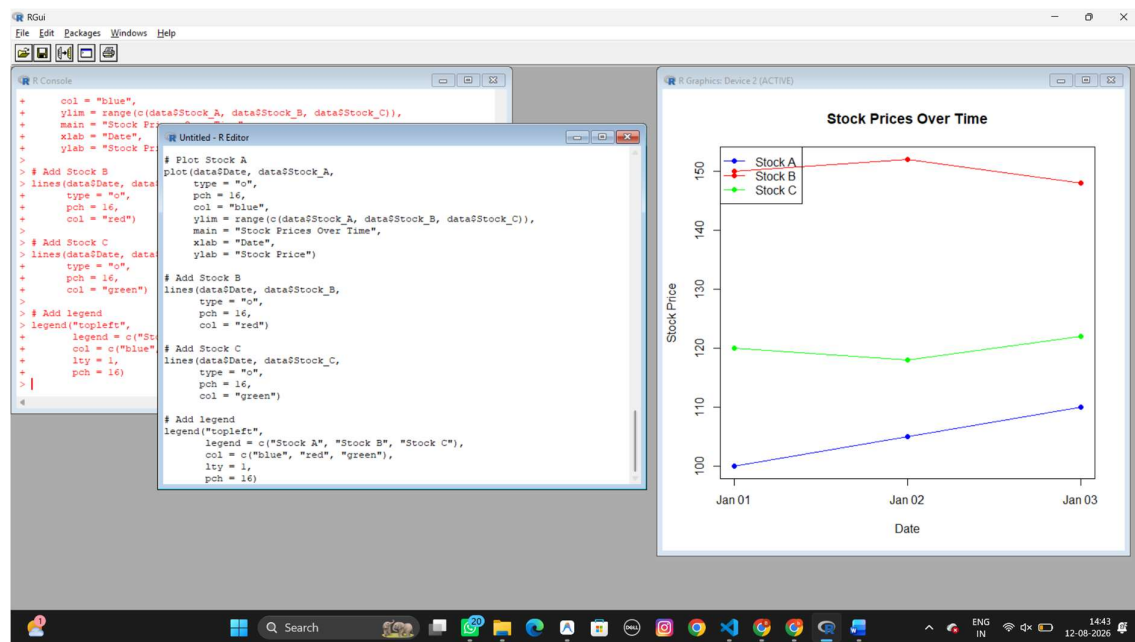
# Display table
print(data)

```

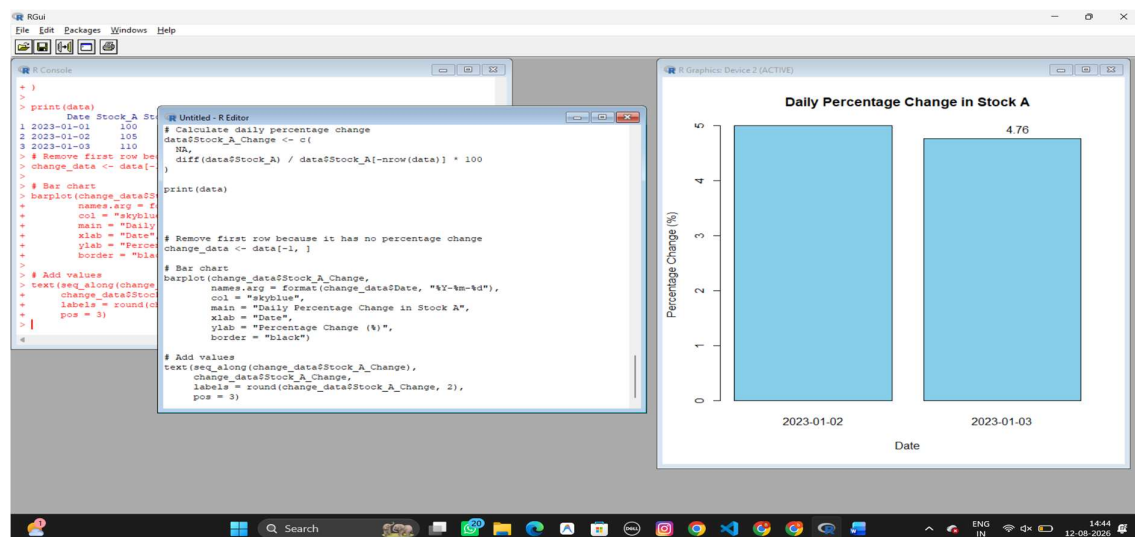
Set 20: Stock Analysis

Dataset: Stock Prices

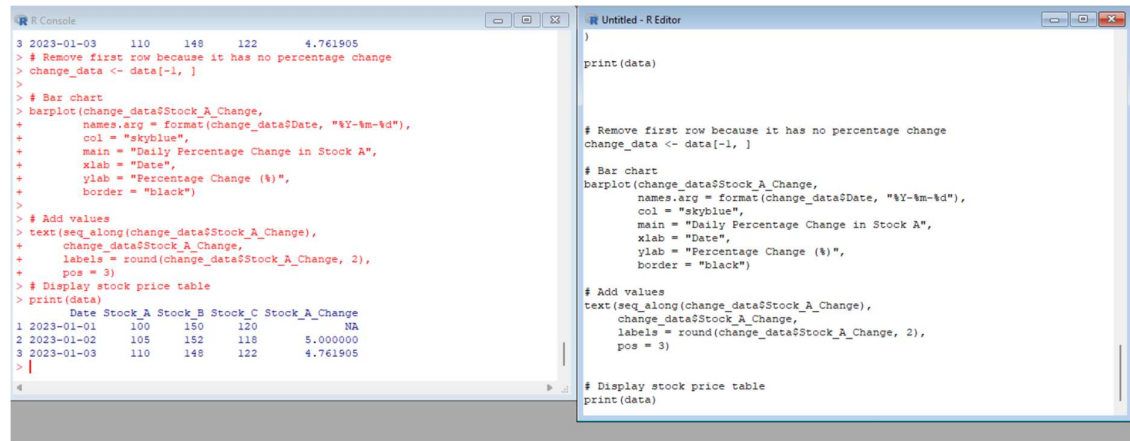
1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.



2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.



3. Build a table to display the stock price data for each company over the given period. Label the table elements.



The image shows two windows from an R environment. The 'R Console' window on the left displays the execution of R code, including a bar plot and a printed table of stock price data. The 'Untitled - R Editor' window on the right shows the source code for the same operations.

```
R Console
3 2023-01-03    110    148    122    4.761905
> # Remove first row because it has no percentage change
> change_data <- data[-1, ]
>
> # Bar chart
> barplot(change_data$Stock_A_Change,
+         names.arg = format(change_data$Date, "%Y-%m-%d"),
+         col = "skyblue",
+         main = "Daily Percentage Change in Stock A",
+         xlab = "Date",
+         ylab = "Percentage Change (%)",
+         border = "black")
>
> # Add values
> text(seq_along(change_data$Stock_A_Change),
+      change_data$Stock_A_Change,
+      labels = round(change_data$Stock_A_Change, 2),
+      pos = 3)
> # Display stock price table
> print(data)
  Date Stock_A Stock_B Stock_C Stock_A_Change
1 2023-01-01    100    150    120          NA
2 2023-01-02    105    152    118    5.000000
3 2023-01-03    110    148    122    4.761905
>

Untitled - R Editor
)
print(data)

# Remove first row because it has no percentage change
change_data <- data[-1, ]

# Bar chart
barplot(change_data$Stock_A_Change,
        names.arg = format(change_data$Date, "%Y-%m-%d"),
        col = "skyblue",
        main = "Daily Percentage Change in Stock A",
        xlab = "Date",
        ylab = "Percentage Change (%)",
        border = "black")

# Add values
text(seq_along(change_data$Stock_A_Change),
     change_data$Stock_A_Change,
     labels = round(change_data$Stock_A_Change, 2),
     pos = 3)

# Display stock price table
print(data)
```